

Research Article

A Game Theoretic Model for Wireless Sensor Networks with Hidden-Action Attacks

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The unattended nature of wireless sensor networks makes them very vulnerable to an adversary's malicious attack. In this paper, we propose to apply game theory into solving the network security problem of wireless network. We explore game theory algorithms to model situation for wireless network with malicious nodes and investigate the attack and detection problem by modeling it as pairwise simultaneous game and spatial structured game. We consider the relationship between the nodes in a wireless sensor network to formulate the game and give the game theory algorithms in detail. We also evaluate the approach with a simulation experiment and analyze the simulation results in detail. We argue that the approach is able to support secure end-to-end communication in wireless sensor networks.

1. Introduction

The unattended nature of wireless sensor networks (WSNs) makes them very vulnerable to an adversary's malicious attacks. An adversary can physically compromise a subset of nodes in a WSN to eavesdrop information. The compromised nodes (or malicious nodes) become *black holes* [1] in the network. Those black holes in a WSN raise hidden-action attacks [2] to reduce the performance of the network or even destroy the network. Therefore, network security is an important issue for WSNs. In a WSN with malicious nodes, there are obvious conflicts between malicious nodes and normal nodes. As a branch of applied mathematics, game theory is concerned with how rational entities make decisions in a situation of conflict, which has been used primarily in economics. Game theory aims to model situations in which decision makers have to make specific actions that have mutual—possibly conflicting—consequences [3]. In the context of WSNs, game theory may be used to form cooperation schemes among entities in a competitive environment, for example, power control, routing, and resource allocation. Game theory can be applied to model the situation where there are malicious nodes in a WSN.

In this paper, we try to explore game theory to model situation for wireless network with malicious nodes and solve

the problem of secure wireless communications by using a game-based approach. The main contributions of this paper are summarized as follows: (1) a formal representation model for simulating a WSN with malicious nodes and hidden-action attacks by using game theory is given; (2) several efficient game-based algorithms are proposed to support reliable and secure wireless communications against the attacks of malicious nodes in the WSN. The remaining of the paper is organized as follows. In Section 2, we first illustrate the system model including the network model as well as the problem statement. In Section 3, we present the game theoretic formulation for a WSN with malicious nodes. Game theory is used to solve the problem of reliable and secure wireless communications against the active attacks of malicious nodes in WSN. Moreover, we present the game theory algorithms in Section 4. In Section 5, we also evaluate the approach with a simulation experiment and analyze the simulation results in detail. Section 6 gives an overview of the related works. Section 7 concludes the paper with an outlook on future research directions.

2. Network Model and Problem Statement

2.1. Network Model. In this paper, we consider a relatively simple WSN. Consider $I = \{1, 2, \dots, k\}$, and assume a WSN

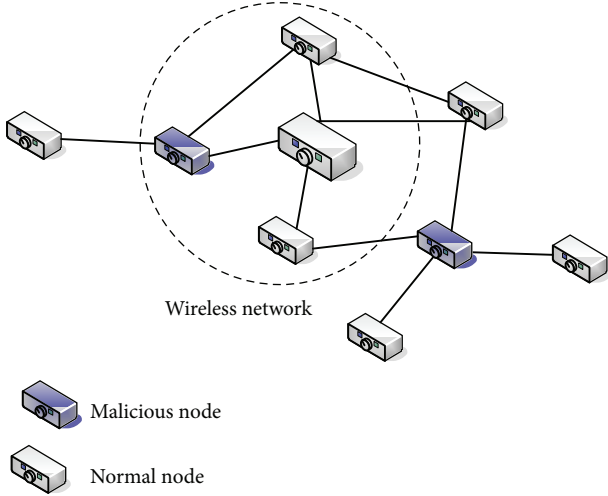


FIGURE 1: A wireless sensor network with malicious nodes.

with k nodes (see Figure 1). Each node has a unique identity i ($i \in I$) in the network. We assume the locations of the nodes are relatively static rather than dynamic. Each node in the network has at least one neighbor node. Each node in the network is battery powered and has limited computation and wireless communication capabilities. Assume the initial energy value of each node in the network is η . Each normal node in the network has four kinds of actions.

- (1) *Forward* means a node sends a data packet to another node in the network.
- (2) *Receive* means a node receives a data packet from another node in the network.
- (3) *Detect* means a node receives a data packet from another node in the network and analyzes the pattern of the packet to find out any abnormality.
- (4) *Sleep* means a node turns off its antenna and does not take any other actions.

Each action (except sleep) of a node will consume certain of energy. Therefore, either a normal node or a malicious node intends to sleep periodically, in order to save energy. An adversary is able to compromise a node or even physically capture a node. Therefore, there are a number of malicious nodes in the network. Assume the number of the malicious nodes is h ($0 < h \ll k$). Each malicious node in the network has four kinds of actions.

- (1) *Forward*.
- (2) *Receive*.
- (3) *Jam* means a malicious node sends an abnormal data packet to another node in the network, in order to block the channel of the node.
- (4) *Sleep*.

Normal nodes do not have knowledge of the instantaneous channel state information (CSI) of malicious nodes, but they know their distribution, and vice versa. Moreover,

we assume that the number of the malicious nodes is much smaller than the number of the sensor nodes in the network; that is, $h \ll k$. It makes sense that a network is out of control if the number of the malicious nodes is very large. Another important assumption is that the type or identity (normal or malicious) of the nodes is unknown or blind. A node only knows the identity of itself.

2.2. Problem Statement. The major task of a normal node in the network is to transmit data to other nodes by routing. However, there is an additional task for normal nodes in a WSN with malicious nodes, that is, to detect the abnormal behaviors of malicious nodes. The major task of a malicious node is to block as many data packets of normal nodes as possible by attacking. A malicious node has two kinds of attack methods, passive methods like eavesdropping and active methods like jamming and DoS attack. In WSN, passive attacks of malicious nodes are hard to be detected, while active attacks have a risk of being detected. In this work, we mainly focus on the positive ones. Malicious nodes, in order to allay suspicions or save energy, only take attacks intermittently.

Moreover, we assume that detecting malicious nodes is an accumulative process to a normal node. Every detecting action of a normal node may gain a piece of evidence. When a normal node allocates enough evidence, it could then perform intrusion detection [4, 5] and locate malicious nodes.

3. Game Theoretic Formulation

In this section, we investigate the attack and detection problem by modeling it as a simultaneous game. In this game, we consider the relationship between every pair of nodes in the network. We try to formulate different cases for this game in detail.

3.1. Pairwise Simultaneous Game. Let us consider a WSN with two nodes, a normal node n and a malicious node m . It is a pairwise simultaneous game between the two nodes, where both players move simultaneously. Each player has four strategies (see Table 1). Here γ is the direct income for a node to send out a data packet successfully; that is, the packet is received by another node. α_F is the probability that a normal node sends out a data packet successfully ($1 > \alpha_F > 0$). α_J is the probability that a malicious node sends out a jamming packet successfully ($1 > \alpha_J > \alpha_F$). It makes sense that jamming is more likely to be successful than forwarding. Here p is a punishment to a detected jamming action of the malicious node, and s is a stimulus to a successful jamming detection of the normal node ($p > 0, s > 0$).

We formulate the energy consumption in the game by some constants. β_1 is the cost to receive a packet (receiving or detecting, $\beta_1 > 0$), and $\Delta\beta_1$ is the additional cost for detecting attacks ($\Delta\beta_1 > 0$). β_2 is the cost to send out a packet (forwarding or jamming, $\beta_2 > 0$), and $\Delta\beta_2$ is the additional cost for jamming because jamming action consumes more energy compared with normal forwarding action ($\Delta\beta_2 > 0$).

TABLE 1: The result of two nodes' game (one normal node versus one malicious node).

	Normal node (n)			
	Forward	Detect	Receive	Sleep
Malicious node (m)				
Forward	$\{-\beta_2, -\beta_2\}$	$\{-\beta_2, -(\beta_1 + \Delta\beta_1)\}$	$\{\alpha_F \gamma - \beta_2, -\beta_1\}$	$\{-\beta_2, 0\}$
Jam	$\{-(\beta_2 + \Delta\beta_2), -\beta_2\}$	$\{-p - (\beta_2 + \Delta\beta_2), s - (\beta_1 + \Delta\beta_1)\}$	$\{\alpha_J \gamma - (\beta_2 + \Delta\beta_2), -\beta_1\}$	$\{-(\beta_2 + \Delta\beta_2), 0\}$
Receive	$\{-\beta_1, \alpha_F \gamma - \beta_2\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, 0\}$
Sleep	$\{0, -\beta_2\}$	$\{0, -\beta_1\}$	$\{0, -\beta_1\}$	$\{0, 0\}$

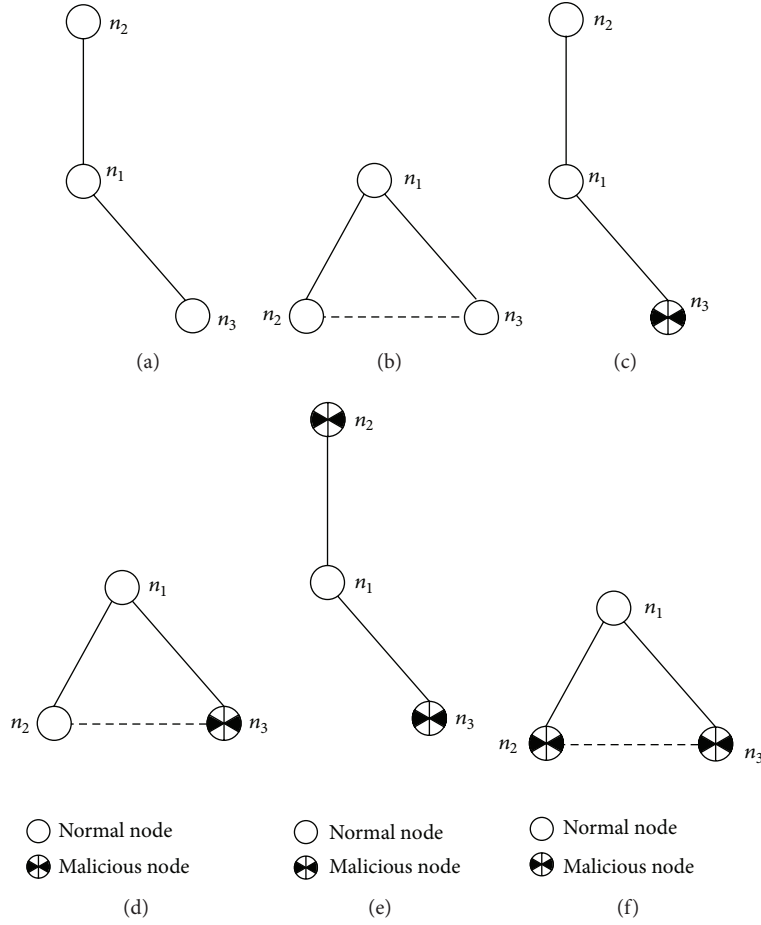


FIGURE 2: Different patterns for local subgames.

As we have mentioned before, nodes do not know the identity of each other in our network model. Even a malicious node is not able to confirm one of its neighbor nodes as its ally. Therefore, there is also a game between two malicious nodes, m_1 and m_2 . The result of two malicious nodes' game is shown in Table 2.

Similarly, there is a game between two normal nodes, n_1 and n_2 . The result of two normal nodes' game is illustrated in Table 3.

3.2. Spatial Structured Game. Let us consider a more complex situation. There are a number of subgames between different pairs of nodes (either normal or malicious) in the whole

network. We could reduce the complete game into a number of two nodes' games. For each two nodes' game, we could get a result by the formulation in Section 2.2 and then synthesize the result of the complete game by the result of all the subgames. Considering a part of a WSN with three nodes (n_1, n_2, n_3), we could get a series of patterns according to the network topology (see Figure 2).

The strategy set of a normal node is $S_{\text{normal}} = \{F, D, R, S\}$, and the strategy set of a malicious node is $S_{\text{malicious}} = \{F, J, R, S\}$. The meanings of F , D , J , R , and S are represented in Table 1. We use $*$ to denote any action in S_{normal} or $S_{\text{malicious}}$. Moreover, we use $-X$ to denote any action except X in S_{normal} or $S_{\text{malicious}}$. The payoff function for n_i in the subgame between n_i and n_j is denoted by u_{n_i, n_j} .

TABLE 2: The result of two nodes' game (one malicious node versus one malicious node).

	Malicious node (m_1)			
	Forward	Jam	Receive	Sleep
Malicious node (m_2)				
Forward	$\{-\beta_2, -\beta_2\}$	$\{-\beta_2, -(\beta_2 + \Delta\beta_2)\}$	$\{\alpha_F\gamma - \beta_2, -\beta_1\}$	$\{-\beta_2, 0\}$
Jam	$\{-(\beta_2 + \Delta\beta_2), -\beta_2\}$	$\{-(\beta_2 + \Delta\beta_2), -(\beta_2 + \Delta\beta_2)\}$	$\{\alpha_J\gamma - (\beta_2 + \Delta\beta_2), -\beta_1\}$	$\{-(\beta_2 + \Delta\beta_2), 0\}$
Receive	$\{-\beta_1, \alpha_F\gamma - \beta_2\}$	$\{-\beta_1, \alpha_J\gamma - (\beta_2 + \Delta\beta_2)\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, 0\}$
Sleep	$\{0, -\beta_2\}$	$\{0, -(\beta_2 + \Delta\beta_2)\}$	$\{0, -\beta_1\}$	$\{0, 0\}$

TABLE 3: The result of two nodes' game (one normal node versus one normal node).

	Normal node (n_1)			
	Forward	Detect	Receive	Sleep
Normal node (n_2)				
Forward	$\{-\beta_2, -\beta_2\}$	$\{-\beta_2, -(\beta_1 + \Delta\beta_1)\}$	$\{\alpha_F\gamma - \beta_2, -\beta_1\}$	$\{-\beta_2, 0\}$
Detect	$\{-(\beta_1 + \Delta\beta_1), -\beta_2\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, 0\}$
Receive	$\{-\beta_1, \alpha_F\gamma - \beta_2\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, -\beta_1\}$	$\{-\beta_1, 0\}$
Sleep	$\{0, -\beta_2\}$	$\{0, -\beta_1\}$	$\{0, -\beta_1\}$	$\{0, 0\}$

Assume that n_1 is always a normal node in this part and there are four cases for the identities of n_2 and n_3 .

In pattern (a) and pattern (b), both neighbors of n_1 are normal. There is a subgame between n_1 and n_2 . There is another subgame between n_1 and n_3 . Due to the broadcast character of WSN, we shall not calculate the cost of a node for more than one time when we compute the payoff of the node that plays subgames with several nodes at the same time.

In these patterns, we have the following exception rules to the game result in Table 3.

- (1) If $u_{n_1, n_2}(F, R) = \alpha_F\gamma - \beta_2$, then $u_{n_1, n_3}(F, *) = 0$, and vice versa (Rule 1).
- (2) If $u_{n_1, n_2}(F, *) = -\beta_2$, then $u_{n_1, n_3}(F, -R) = 0$ and $u_{n_1, n_3}(F, R) = \alpha_F\gamma$, and vice versa (Rule 2).
- (3) If $u_{n_1, n_2}(D, *) = -(\beta_1 + \Delta\beta_1)$ or $u_{n_1, n_2}(D, *) = -\beta_1$, then $u_{n_1, n_3}(D, *) = 0$, and vice versa (Rule 3).
- (4) If $u_{n_1, n_2}(R, *) = -\beta_1$, then $u_{n_1, n_3}(R, -F) = 0$, and vice versa (Rule 4).

In pattern (c) and pattern (d), n_1 has only one malicious neighbor (n_3), and the its other neighbor (n_2) is normal. Besides the constraint in patterns (a) and (b), there is one more constraint for the three nodes in these two patterns. In pattern (c), the data that flows from n_2 to n_1 is affected by n_3 . Therefore, when we compute the subgame between n_1 and n_2 , we should consider this constraint. In pattern (d), there is one more edge between n_2 and n_3 . That is, n_3 is a common neighbor of n_1 and n_2 . It means that n_3 has effects on both n_1 and n_2 , or both the data that flows from n_1 to n_2 and the one from n_2 to n_1 are affected by n_3 .

In pattern (c), we have the following exception rules to the game results in Tables 1 and 3.

- (1) Rule 1.
- (2) Rule 2.

- (3) If $u_{n_1, n_2}(D, *) = -(\beta_1 + \Delta\beta_1)$ or $u_{n_1, n_2}(D, *) = -\beta_1$, then $u_{n_1, n_3}(D, *) = 0$, and vice versa (Rule 5).
 - (4) If $u_{n_1, n_2}(R, *) = -\beta_1$, then $u_{n_1, n_3}(R, -F) = 0$, and vice versa (Rule 6).
 - (5) If $u_{n_3, n_1}(J, R) = \alpha_J\gamma - (\beta_2 + \Delta\beta_2)$, then $u_{n_2, n_1}(F, R) = -\beta_2$ (Rule 7).
- In pattern (d), we have one more exception rule.
- (6) If $u_{n_3, n_1}(J, R) = \alpha_J\gamma - (\beta_2 + \Delta\beta_2)$, then $u_{n_1, n_2}(F, R) = -\beta_2$ (Rule 8).

In pattern (e) and pattern (f), both neighbors of n_1 are malicious. Similarly, with pattern (c) and pattern (d), a malicious node will affect the subgame of its neighbor. In pattern (e), the data that flows from n_2 to n_1 is affected by n_3 , while the data that flows from n_3 to n_1 is also affected by n_2 . In pattern (f), both the data flow from n_1 to n_2 and the one from n_2 to n_1 are affected by n_3 , while both the data that flows from n_1 to n_3 and the one from n_3 to n_1 are affected by n_2 .

In pattern (e), we have the following exception rules to the game results in Table 2.

- (1) Rule 1.
 - (2) Rule 2.
 - (3) If $u_{n_1, n_2}(R, *) = -\beta_1$, then $u_{n_1, n_3}(R, R) = 0$ and $u_{n_1, n_3}(R, S) = 0$, and vice versa (Rule 9).
 - (4) If $u_{n_2, n_1}(J, R) = \alpha_J\gamma - (\beta_2 + \Delta\beta_2)$, then $u_{n_3, n_1}(F, R) = -\beta_2$, and vice versa (Rule 10).
- In pattern (f), we have one more exception rule.
- (5) If $u_{n_2, n_1}(J, R) = \alpha_J\gamma - (\beta_2 + \Delta\beta_2)$, then $u_{n_1, n_3}(F, R) = -\beta_2$, and vice versa (Rule 11).

If we take the whole network as an undirected graph, each edge denotes a subgame between two nodes. We could compute the game result of each edge in the graph by traversing the graph with a certain traversal algorithm (e.g., BFS or DFS).

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Input: A wireless node  $n_0$ .
Output: A payoff value  $p_0$ .
get the neighbour nodes of  $n_0$  as  $NB = \{n_1, n_2, \dots, n_k\}$ 
for each node  $n_i$  in  $NB$  ( $i = 0, 1, \dots, k$ )
    perform a two nodes' game between  $n_0$  and  $n_i$ 
end loop
construct a many-to-many game for  $n_0$  and  $NB$ 
compute payoff  $p_0$  for  $n_0$  under the restriction of the rules
in Section 3.2
return  $p_0$ 

```

ALGORITHM 1: The game algorithm for a wireless node.

4. Game Theory Algorithms

We could formulate the game for a WSN given in Section 3 by the following algorithms. First, let us consider the case for one specific wireless node (either normal node or malicious node). Algorithm 1 is proposed to compute the payoff of a specific wireless node.

The algorithm is trivial. The payoff of a specific node is impacted by the roles of its neighbors. In order to compute the payoff of a specific node, we have to consider the relationship between the node and each of its neighbors.

Then we could compute the payoffs of the whole network based on Algorithm 1. The algorithm for computing global payoffs is given as follows.

Algorithm 2 is also easy to be understood. The process of computing global payoffs is reduced to traversing an undirected graph by BFS. For each node, we just perform a single-node payoff computation, and then we get the information for the whole network. To a given WSN, we shall perform a series of computations to reach convergence.

5. Simulation and Evaluation

5.1. Simulation Setting. In this section, we construct simulation to evaluate the performance of the proposed approach. We consider a WSN with N players (wireless nodes). Player i is represented as a vertex v_i of a graph $G(V, E)$, with $v_i \in V$. An interaction between two players i and j is represented by an undirected edge $e_{ij} \in E$. The number of neighbors of player i is the degree k_i of vertex v_i . The average degree of the network is denoted by $\langle k \rangle$. The terms vertex, individual, participant, and player are used interchangeably in this section, likewise for edge, interaction, and link. Each malicious node can take one of the four strategies: $\{F, D, R, S\}$, while each normal node can take one of the four strategies: $\{F, J, R, S\}$.

The *BA scale-free* network [6] is adopted to represent the population structure of our WSN, which is constructed according to the “growth” and “preferential attachment” mechanisms. Starting from m_0 fully connected nodes, a new node with m ($m \leq m_0$) edges is added to the system at every step. The new node links to m different nodes by a “preferential attachment” mechanism. The probability of connecting to an existing node i is proportional to its degree;

that is, $p_i = k_i / \sum_j k_j$, where j runs over all existing nodes and k_i is the degree of node i . After t time steps, this algorithm produces a graph with $N = t + m_0$ vertices and mt edges.

Let $S(t) = (s_1(t), s_2(t), \dots, s_N(t))$ denote a configuration of the population strategies $s_i(t) \in \{F, D, J, R, S\}$ at time step t ; the global synchronous system dynamics leads to $S(t+1)$ by simultaneously updating all the players' strategies according to the chosen rule, such as Fermi update rule [7]. Here, by synchronous, we mean that player's strategy will not change (even if it had already changed during its strategy updating process at time step t) until all the other $N-1$ players complete their strategy updating processes.

The game process is the same as the standard evolutionary game. At each step, all nodes are synchronously updated according to a strategy update rule. Note that in realistic WSNs, there may be environmental noise, which influences individuals' decisions (e.g., strategy mutation). Therefore, we adopt the Fermi updating rule, which considers environmental noise. When player i updates its strategy, it will first select a neighbor j out from all its k_i neighbors at random and then adopt j 's strategy with the probability $H_{i \rightarrow j} = 1/(1 + e^{(P_i - P_j)/\kappa})$.

Here, κ characterizes the environmental noise, including bounded rationality, individual trials, and errors in decision; $\kappa \rightarrow \infty$ leads to neutral (random) drift whereas $\kappa \rightarrow 0$ corresponds to the imitation dynamics, where player j 's strategy replaces player i 's whenever $P_j > P_i$. For finite value of κ , the smaller the κ is, the more likely the fitter strategy is to replace the less fit one; thus the value of κ indicates the intensity of selection.

Finally, Tables 1–3 describe the payoffs of normal nodes and malicious nodes; Figure 2 depicts all the six exception rules in the network.

5.2. Experiment Results. In the following simulations, we study the effect of β_1 (cost of receiving a package), β_2 (cost of sending out a package), $\Delta\beta_1$ (additional cost of detecting attacks), $\Delta\beta_2$ (additional cost of jamming), p (punishment to a detected jamming action of the malicious node), and s (stimulus to a successful jamming detection of the normal node). α_F is set to 75%, and α_J is set to 100% (i.e., whenever a malicious node decides to jam the network, it will succeed). Besides, γ the direct income for a node to send out a data packet successfully is set to unit payoff. All the simulations were carried out on a *BA scale-free* network with network size $N = 1000$ and $m = m_0 = 4$; therefore, the average degree $\langle k \rangle = 8$. Initially, normal nodes and malicious nodes are randomly distributed among the population with probability of 90% and 10%, respectively. In addition, for normal nodes, the four strategies, that is, F, D, R , and S , are assigned the same probability, that is, 25%; for malicious nodes, the four strategies, that is, F, J, R , and S , are also assigned the same probability. The equilibrium frequencies of malicious jamming nodes are obtained by averaging over 3000 generations after a transient time of 10,000 generations. Each piece of data is averaged over 1000 runs on 1000 different networks.

Input: A WSN with a collection of nodes $N = \{n_0, n_1, \dots, n_k\}$, a source node n_0 , and a collection of malicious nodes $M = \{m_1, m_{i+1}, \dots, m_h\}$, where $n_0 \in N, M \subset N$.
Output: A collection of payoff values $P = \{p_0, p_1, \dots, p_k\}$ for N , where p_i is the payoff of n_i ($i = 0, 1, \dots, k$).
Assign an initial action to each n_i in N and the action of n_0 is *Forward*
Set n_0 as the root node
Perform BFS by starting with n_0
for each intermediate node n_i in BFS
 compute payoff for n_i by using **Algorithm 1**
 get p_i for n_i
end loop
Return $P = \{p_0, p_1, \dots, p_k\}$

ALGORITHM 2: The game algorithm for a WSN with malicious nodes.

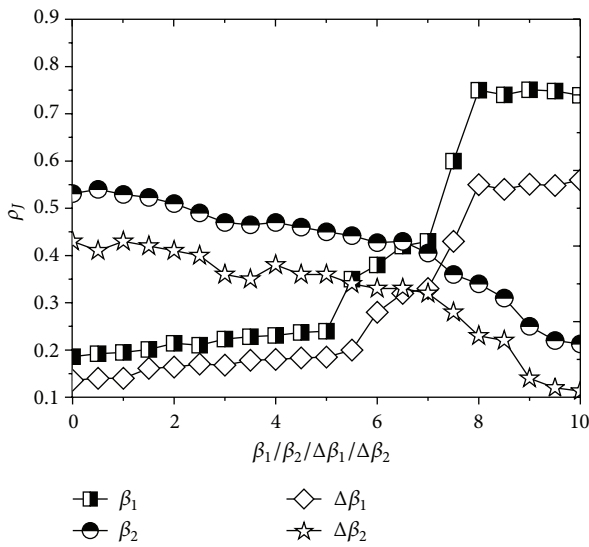


FIGURE 3: Effect of package-receiving cost, package forwarding cost, jamming detecting cost, and jamming cost on the proportion of malicious jamming nodes.

5.2.1. Action Cost Effect. Figure 3 shows the relationship between network action (F, D, J, R , and S) cost and ρ_J , where $\rho_J = N_m^{J*}/N_m$ is the ratio of the number of malicious nodes that take the jamming action N_m^{J*} to the total number of malicious nodes N_m in the equilibrium state. As previously referred, the initial proportion of malicious nodes $N_m/N = 10\%$ and of N_m^F (number of initial malicious forwarding nodes), N_m^R (number of initial malicious receiving nodes), N_m^J (number of initial malicious jamming nodes), and N_m^S (number of initial malicious sleeping nodes) is assigned 25%, respectively. We can see that by increasing package-receiving cost or additional detecting cost, ρ_J increases; however, when increasing package-sending cost or additional jamming cost, ρ_J decreases. We argue that this result is expected. It makes sense that normal nodes incline to perform package sending rather than detecting when the detecting cost increases. As a result, malicious nodes are more likely to perform jamming because there are not enough normal nodes detecting their

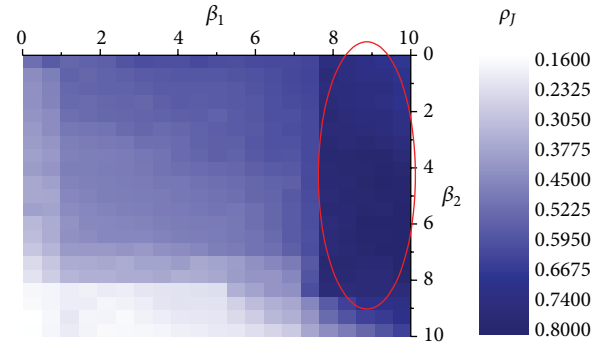


FIGURE 4: Coaction effect of forwarding and receiving on the proportion of malicious jamming nodes.

attacks. Moreover, malicious nodes incline to be (or pretend to be) “normal” or when the cost of attacks increases.

5.2.2. Coaction Effect. To further explore the effect of network action cost, we jointly study the coaction of forwarding and receiving. The result is shown in Figure 4. The value of ρ_J is represented in the form of color depth. As Figure 3 indicated, malicious nodes that take the jamming action will emerge when β_1 is large and β_2 is small. On the other hand, when β_1 is small and β_2 is large, the self-evolved network can inhibit hidden-action attack, for example, jamming, effectively. Figure 4 just confirms our prediction. As we can see, the deepest blue area locates in the northeast corner of the axes and the lightest blue area locates in the southwest corner of the axes. We can also find that β_1 performs a bigger weight than β_2 on the equilibrium proportion of malicious nodes that take the jamming action. We also find that when β_1 is large enough, variation of β_2 has little effect on ρ_J (see the red-circled area in Figure 4). This result is in accordance with Figure 3, where the $\beta_1 - \rho_J$ curve is steeper than the $\beta_2 - \rho_J$ curve, indicating that change of β_1 is much more influential.

5.2.3. Stimulation and Punishment Effect. Next, we explore the impacts of detecting stimulation s and jamming punishment p on ρ_J . The result is shown in Figure 5. It is indicated

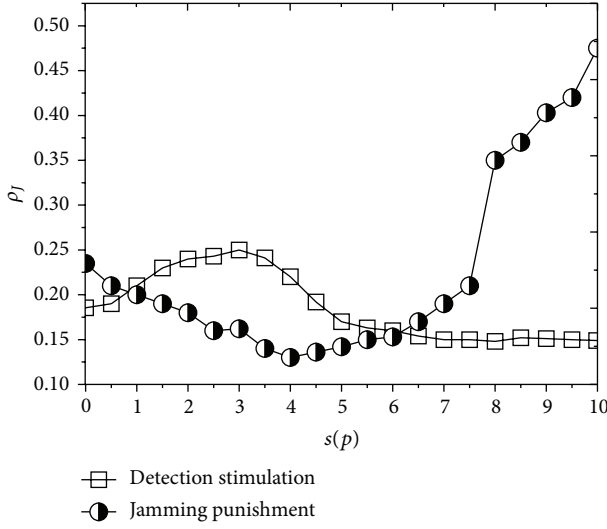


FIGURE 5: Effect of detecting stimulation and jamming punishment on the proportion of malicious jamming nodes.

that, for detecting stimulation, as s increases, ρ_J firstly increases to a local maximum and then decreases to a stable value. For jamming punishment, as p increases, ρ_J firstly decreases to a local minimum and then increases to about 50%. We argue that both of two curves are just as we expected. For detecting stimulation, when s is small, it is not enough to encourage normal nodes to detect malicious jamming nodes until it reaches a threshold value, after which more normal nodes joined in for detection, and thus ρ_J decreases. However, when s is large, increase of detecting stimulation makes no sense. Because the number of malicious jamming nodes is small, the total detecting payoff is limited, which is unable to attract more normal nodes to join and perform detecting. For jamming punishment, when p is small, the malicious nodes that take the jamming action are efficiently constrained. However, when p is large, the punishment cost exceeds the income for punisher; thus the number of normal nodes that take the detecting action decreases and results in the increase of ρ_J .

Similarly, we also study the joint effect of s and p on the proportion of malicious nodes that take jamming action. The result is given in Figure 6. According to the result of Figure 5, a large ρ_J will appear when p is large and s is around 3. However, as Figure 6 depicts, the maximum of ρ_J has an apparent drift towards larger value of s (see the blue-circled area). This can be explained as follows: when the jamming punishment p is large, normal nodes are unwilling to perform detecting, because the punishment cost exceeds the income for punisher. Therefore, these normal nodes will wait until the detection stimulation increases to an acceptable threshold. The orbit of drift is shown as an arrowed curve in Figure 6. The black-circled area indicated the smallest ρ_J , which is also in accordance with Figure 5. Like that in Figure 4, here we argue that the jamming punishment p performs a bigger weight than detecting stimulation s on

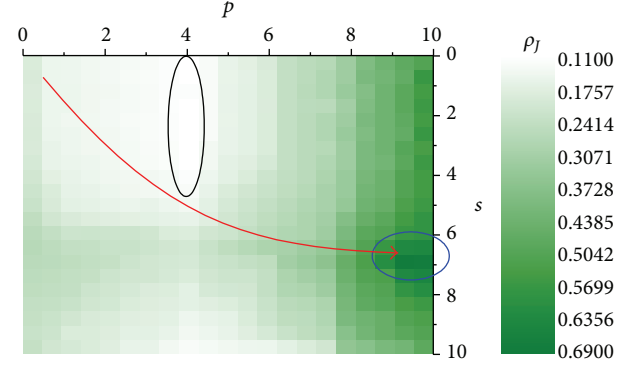


FIGURE 6: Joint effect of detecting stimulation and jamming punishment on the proportion of malicious nodes taking the jamming action.

the equilibrium proportion of malicious nodes that take the jamming action.

6. Related Works

Applying game theory into improvement the QoS of WSN or wireless network is a new research direction, especially the games for the security in WSN or wireless network. There have been a few research efforts in this field. However, there are still many issues worthy of further exploration.

Kodialam and Lakshman in [8] model a zero-sum game between the intruder and service provider of the network. The objective of the intruder is to inject a malicious packet in the network at some node with node t as the target. The intrusion is successful when the packet reaches the target and unsuccessful when it does not. To protect the nodes from the attack, the service provider is allowed to sample the packets flowing through the links on the network. The optimal solution for this game is the min.-max. optimal solution, which is the Nash equilibrium for a zero-sum game. Agah et al. in [9] study a game formulation at the routing layer of a wireless network between malicious nodes that do not forward incoming packets and an intrusion detector residing at the base station. They model this scenario as a repeated game, where the IDS uses the history of nodes' collaboration to determine paths comprising malicious nodes. The proposed protocol for the repeated game shows a correlation between network size and successful intrusion detection, where detection success rate increases with higher percentage of malicious nodes. Kamhoua et al. in [10] investigate a situation that it is cost effective to freely participate in the security mechanism or protect its privacy depending for each node on the fact that if that node believes or trusts that all other nodes or at least a minimum number of other nodes will do the same. They model a trustable dilemma for autonomous multihop networks by using the mathematical framework of game theory and evolutionary game theory. The well-known stag hunt game is used as their basic game model. They present the interconnection between cooperation, trust, privacy, and security in a network. However, they only

consider the game for normal nodes in the network rather than the game between normal nodes and malicious nodes.

Khouzani et al. in [11] develop a zero-sum dynamic game model and investigate the structural properties of the saddle-point strategies. They consider a game with two players (normal nodes and malicious nodes) and three different states with normal nodes. Their research efforts show that saddle-point strategies are simple threshold-based policies, and hence a robust dynamic defense is practicable. They assume there is no competition among normal nodes or malicious nodes in the network, which is not feasible to many situations in WSNs. Sagduyu et al. in [12] present a class of jamming games played at the MAC layer of wireless network among a set of transmitters and jammers. They address the incomplete information or uncertainty in games compared with existing works. The equilibrium strategies resulting from the jamming games characterize the expected performance under DoS attacks and motivate robust network protocol design to support secure wireless communications. Although they consider the conflicts among malicious nodes to be incomplete information, they have not taken the conflicts of normal nodes into consideration. Jaramillo and Srikant in [13] discuss the problem that packet collisions and interference may make cooperative nodes appear selfish sometimes, generating unnecessary and unwanted punishments in wireless ad hoc networks. They present a robust mechanism to imperfect measurements, which is collusion resistant and can achieve cooperation among nodes. However, their method does not satisfy the situation where there are malicious nodes in the network. Yan et al. in [14] proposed a penalizing mechanism to prevent the noncooperative selfish behavior of decreasing the contention window without permission based on repeated game theory for WSNs. A Contention Window Select Game is defined, and in this game each sensor node selects its own contention window to control the access probability.

Moreover, there are some research efforts about using other evolutionary algorithms rather than game theory to solve the security problem in WSNs or wireless networks. For example, Alrajeh et al. in [15] present an adaptive secure routing protocol based on bioinspired technique termed as ant colonization for WSNs. Their approach is able to select optimal paths from source to destination by ensuring adaptability, robustness, and security. However, their approach cannot suppress hidden-action attacks in WSNs. Hortos in [16] presents a cross-layer approach to WSN protocol design by applying a bioinspired evolutionary computational method to the functions of each protocol layer to improve the intrusion detection identification (IDID) performance. Genetic algorithms and ant colony optimization are used to solve the problem in the proposed approach.

Although many existing approaches have taken game theory to solve the security problem in WSNs or wireless networks, compared with existing works in this field, our approach considers the relationship between nodes in a WSN and gives a formal theoretic model for the game. We discuss different patterns in this game and represent them as rules to generate a more complete game model. We argue that

such a representation model is relatively lacking in existing research efforts. Moreover, the algorithms proposed in this paper are not very complex and easy to be implemented in resource-constrained WSNs, compared to the existing approaches. In general, our approach is able to support secure wireless communication in WSNs with malicious nodes.

7. Conclusion

In this paper, we mainly present an approach to apply game theory into solving the network security problem of WSN. We try to explore game theory algorithms to model the situation for WSN with malicious nodes, in order to support reliable and secure wireless communications against the attacks of malicious nodes in the network. We consider the relationship between the nodes in a WSN to formulate the game and illustrate the game theory algorithms in detail. In order to verify the proposed approach, we also evaluate it with a simulation experiment and analyze the simulation results in detail. We argue that the approach is able to support secure end-to-end communication in WSNs with malicious nodes.

Future works may include (1) improving the efficiency of the algorithms to support different kinds of attacks rather than jamming, (2) evaluating the cost of the game theory algorithm for WSN, and (3) considering a more complex WSN model to evaluate the approach.

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